Aarush Ram Anandh

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EDUCATION

Ashoka University, Sonipat **2024–2028**
B.Sc. (Hons.) Mathematics and Computer Science

Relevant coursework: Real Analysis, Linear Algebra, Probability and Statistics, Statistical Inference, Machine Learning, Algorithms, Discrete Mathematics; currently pursuing Algebra I, Complex Analysis, Artificial Intelligence, Reinforcement Learning and Symbolic Logic.

RESEARCH EXPERIENCE

Constrained Image-Synthesis Lab, Ashoka University **Sep 2025–Present**
Research Fellow

- Improved colour fidelity in latent diffusion models, achieving $\Delta E_{00} < 1$ and approximately **4× better colour accuracy** than evaluated open-source baselines; published a public preprint.
- Developing **Controlled Generalised Sliced Wasserstein (CGSW)**, a controllable optimal-transport framework with control-conditioned probes, projections and slice aggregation. Abstract submitted to ICLR 2027.
- Studying metricity, admissibility, stability, minimality and optimal probe/readout construction; implementing computational methods in **C++** and machine-verified specifications and proofs in **Lean**.
- Built an image-processing pipeline that generated **20,000+** training inputs for lab use. [GitHub]

Koita Centre for Digital Health, Ashoka University **May 2026–Present**
Research Intern (part-time)

- Developed preprocessing and windowed analysis for longitudinal heart rate, HRV, sleep, SpO_2 and activity data.
- Explored sequential modelling and change-point methods for patient-state analysis of noisy physiological signals.

PUBLICATION & SELECTED RESEARCH

Public preprint – colour-accurate latent diffusion **2026**
Colour-preserving methods for latent diffusion; arXiv:2601.17259.

Hyperspectral Waste Analysis **2026–Present**
Working with a dataset of **hyperspectral image cubes**, each with 224 spectral bands, and masks across seven waste categories. Constructing object-level spectral measures to study class separability using sliced and controlled optimal-transport distances.

ADDITIONAL TECHNICAL & TEACHING EXPERIENCE

Avartan Labs **May–Jul 2026**
Product Engineering / Machine Learning Intern

- Developed an RT-DETR-based detection, classification and tracking pipeline for **18 waste-material classes** in real-time sorting.
- Investigated focal loss, logit adjustment, class-specific thresholds and targeted fine-tuning to improve rare-class performance; evaluated detector failure modes with crop-based refinement.
- Set up and ideation of an a **near-infrared (NIR) waste-segregation system** for visually similar plastics, handling parts procurement and running preliminary experiments.

Lodha Genius Programme, Ashoka University **Jun–Jul 2025**
Teaching Assistant

- Supported four faculty members in an IUCAA telescope workshop and an IoT workshop for **200+ students**.

Yulu Bikes **Apr–May 2023**
Research / Engineering Project

- Modelled electric-vehicle dynamics in **MATLAB/Simulink**, testing the effect of vehicle parameters and operating conditions on system performance.

TECHNICAL SKILLS

Programming: Python, C++, C, OCaml, Bash **Machine learning:** PyTorch, Diffusers, TensorFlow
Formal methods: Lean **Research areas:** Optimal transport, diffusion models, computer vision, time-series analysis
Scientific computing: MATLAB/Simulink, OpenFOAM